7. HORIZONTAL CURVES

As already stated, linear sections of the route are named alignments, while the curved sections that connect alignments to each other are named **curves**. Curves are used when direction changes are needed on the route axis.

7.1 Types and properties of the horizontal curves

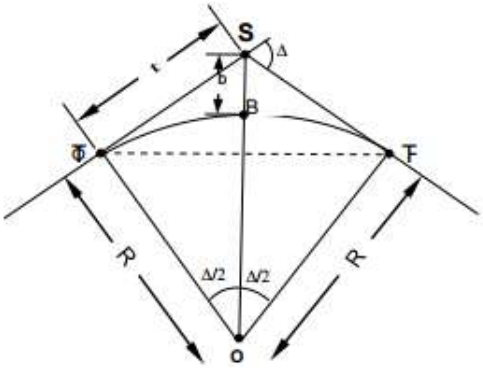
There are some reasons why the route changes its direction. The first reason is the topography of the region. On mountainous or undulating terrain, due to the elevation difference between the necessary points, the direction of the route may change often. On flat terrain, direction change or the need for a horizontal curve is less. However, even if there is no need for one because of the topography, horizontal curves are also used on flat terrain due to the decrement of monotonousness. On very long alignments, the drivers' attention is distracted. On night drives, dazzle occurs because of the headlamp lights of vehicles approaching from the opposite direction. Similarly, if the drive is in the east-west direction during sunrise and sunset, dazzle occurs due to the sunlight. To decrease these negative effects, alignments longer than 8-10 km are not desired. If long alignments are inevitable, high-radius curves are placed within the alignments to decrease their length.

While designing the horizontal curves, the safety, the capacity, and the drive comfort standards have to be fulfilled. On erroneously designed horizontal curves, capacity and the level of service will decrease due to the decrease in the operating velocity. Many accidents may occur as vehicles are exposed to skidding and rollover effects. On the other hand, erroneously designed horizontal curves decrease comfort, result in worse driving conditions, and increase the operating cost of vehicles.

Because of the reasons stated above, the horizontal curves are the critical sections on a highway and should be analyzed thoroughly. Horizontal curves are usually made of circular arcs. When traveling with higher velocities on a horizontal curve, from entering the curve from an alignment, passengers on the vehicles are affected by skidding. With this effect, the trip comfort decreases. To preserve the trip comfort at a certain level, transition curves are placed between an alignment and a curve. This way, the effect of the centrifugal force is decreased. There are three types of horizontal curves. They can be listed as **simple**, **compound,** and **inverse** horizontal curves.

7.1.1 Simple horizontal curves

Simple horizontal curves are used to connect two alignments. There are **five** fundamental elements of a horizontal curve, as seen in the figure below. They can be listed as the **curve angle** (Δ), **radius** (R), **tangent length** (t), **curve length** (L), and **external distance** (b).



The point on which the two alignments are intersected is named the **point of intersection** (S). The external angle between the alignments is named the **curve angle** (Δ). The distance of the tangent points of the circle arc with the alignments to the center of the curve is named the **tangent length** (t). The two tangent lengths of a horizontal curve are equal due to the geometry. The first tangent point is named the **point of curvature** (Φ) (on the back tangent), and the second tangent point is named the **point of tangency** (Ŧ) (on the forward tangent). The length of the circle arc is named the **curve length** (L). The curve length is calculated using the circle arc formula. The perpendiculars drawn from the tangents intersect at the center point (O) on **radius** (R) distance. The midpoint of the horizontal curve is named the **bisector point** (B). The distance of the bisector point to the point of intersection is named the **external distance** (b). The line that connects the center of the horizontal curve to the point of intersection divides the curve angle into two equal angles.

$$t = R \quad tg(\frac{\Delta}{2}) \qquad L = \frac{2 \cdot \pi \cdot R}{360} \cdot \Delta^{\circ}$$

If the angle is given as grads,

$$L = \frac{2 \cdot \pi \cdot R}{400} \cdot \Delta^{g}$$

$$\cos(\frac{\Delta}{2}) = \frac{R}{(R+e)} \rightarrow e = \frac{R}{\cos(\frac{\Delta}{2})} - R$$

$$e = R.(\frac{1}{\cos(\frac{\Delta}{2})} - 1) \qquad e = R \cdot (\sec(\frac{\Delta}{2}) - 1)$$

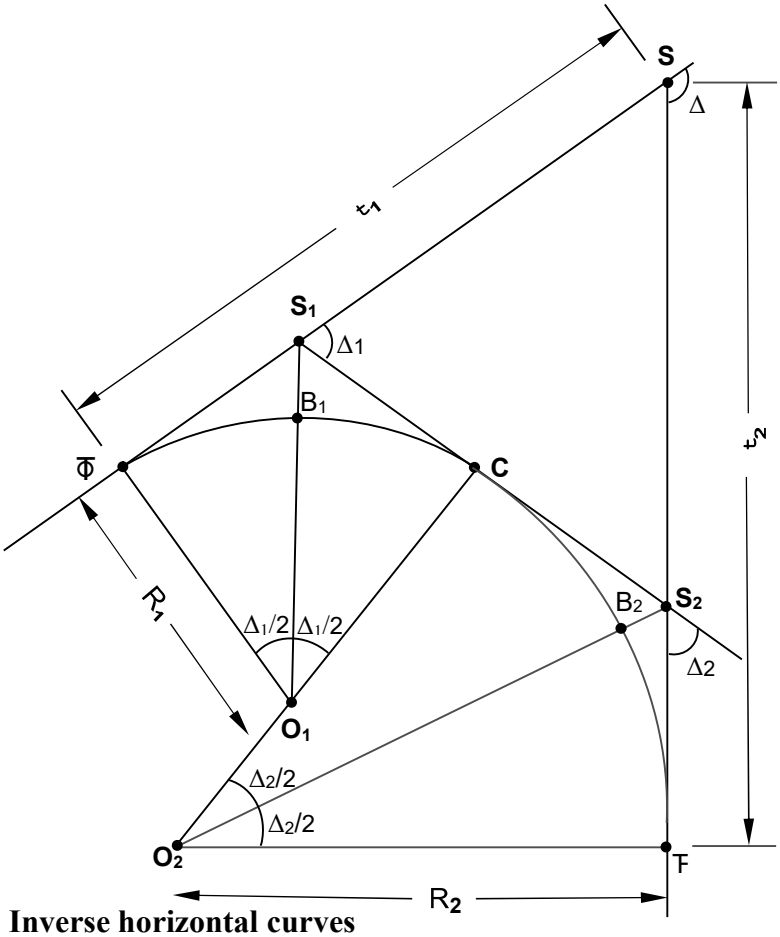
To provide the sight distance, comfort, aesthetics, and safety on horizontal curves, the length of the curve should be sufficient. If the stop-sight distance can not be provided, the curve length should not be shorter than the distance that the vehicle can take in 2-3 seconds while traveling with the project velocity. The minimum curve lengths that can be applied to the design are given below.

V_p (km/h)	L_{min} (m)
50	30
60	35
70	40
80	45
90	50
100	55
110	60
120	65

7.1.2 Compound horizontal curves

They are horizontal curves on the same side of a common tangent having different radii, as seen in the figure. The PT of the first curve and the PC of the second curve are the same point. They are used on special conditions that limit the route design. Especially the topographically compulsory sections to pass on rural highways, natural obstacles that increase the cost and construction limits of urban highways may force the use of conjoint horizontal curves.

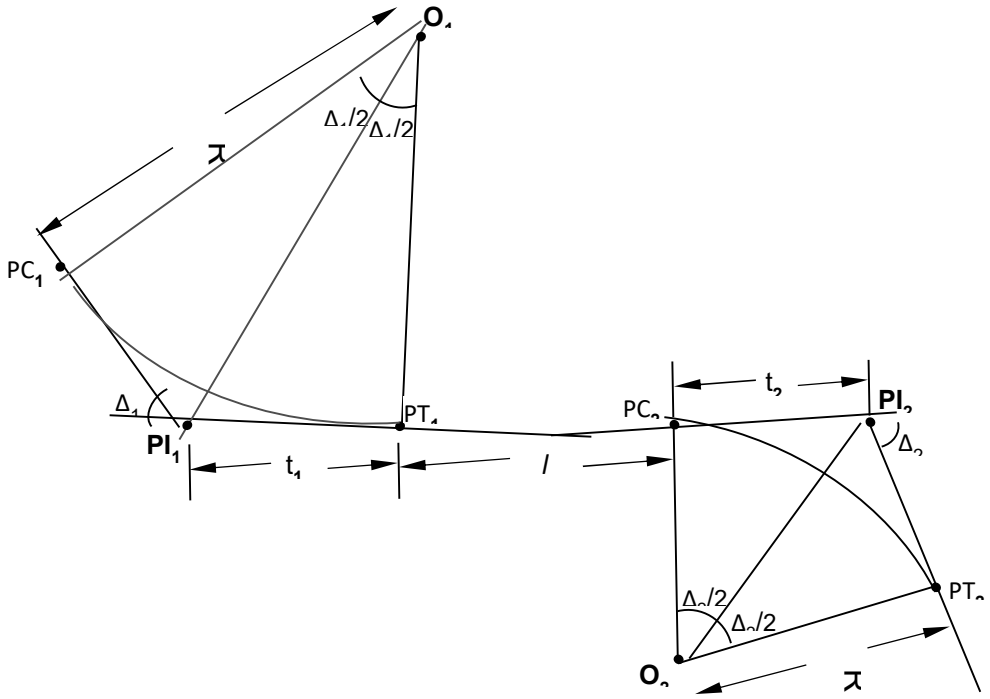
If a compound horizontal curve is used, the bigger curve radius has to be at least 1,5 times the smaller radius. In other words, $R_2 \leq 1{,}5\ R_1$. Compound horizontal curves are determined with seven characteristic components: Δ, Δ_1, Δ_2, R_1, R_2, t_1, and t_2. If at least four of them are known and at least two of them are lengths, the other three components can be calculated.



7.1.3 Inverse horizontal curves

They are composed of two circle arcs placed on each side of a common tangent. As the centers of the curves are in opposite directions, they are named as the **inverse horizontal**

curves. The calculation of each curve is made separately. However, especially on low-standard highways, due to the topographical constraints, if the alignment between the two curves is insufficient, calculations should be made taking both curves into account. It is hard to apply the superelevation and the transition curve on a limited alignment. For this reason, the distance between the PT of curve 1 and the PC of curve 2 has to be at least 60 m. The curve radius should be selected in a way that provides the required distance.

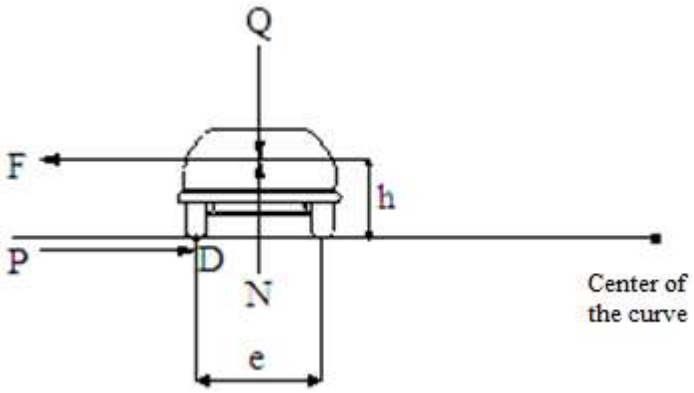


7.2 Vehicle stability on horizontal curves

A vehicle entering a horizontal curve from an alignment moves under the effect of some forces. Although these forces are basically the same, they differ whether there's a slope on the road or not.

7.2.1 Stability on the road without a transverse slope

The forces affecting the situation without a slope are the force composed by the weight of the vehicle, the normal force reacting against it, the latitudinal friction force affecting the contact points of wheels with the rolling surface and through the center of the curve, and the centrifugal force affecting through the outer of the curve as seen on the figure below.



Q : The weight of the vehicle
N : The normal force affecting the opposite side of the vehicle's weight
P : Latidunal friction force
F : Centrifugal force
e : The distance between the wheels of the vehicle (from axis to axis)
h : The distance between the center of gravity of the vehicle and the rolling surface
The centrifugal force that skids the vehicle first, then rollovers it according to the velocity can be written using the basic principle of the dynamics as:

$$F = m \frac{v^2}{R} = \frac{Q}{g} \frac{v^2}{R}$$

m : The mass of the vehicle (kg)
Q : The weight of the vehicle (N)
g : Gravitational acceleration (m/sec^2)
v : Velocity of the vehicle (m/sec)
R : Radius of the horizontal curve (m)

The velocity that causes the skid

If the vehicle that is moving on a curve without a slope, exceeds some critical velocity values, it encounters skid and the rollover effects. When the vehicle exceeds the first critical velocity value, it skids. The velocity that causes the skid can be found by writing the equilibrium conditions. The equilibrium conditions:

$$F = P \qquad Q = N$$

The frictional force is the function of the normal force affecting the object and the frictional coefficient and can be written as:

$$P = \mu_e \quad N$$

If the Q term is written instead of the N term on the second equilibrium condition, the friction force becomes:

$$P = \mu_e \quad Q$$

Then, the latitudinal friction coefficient can be written depending on the vehicle weight:

$$\mu_e = \frac{P}{Q}$$

If the centrifugal force is written instead of the frictional force on the first equilibrium condition, the latitudinal friction coefficient can be written as:

$$\mu_e = \frac{\frac{Q}{g}\frac{v^2}{R}}{Q} = \frac{v^2}{g\ R}$$

If the velocity of the vehicle is written as km/h (v=V/3,6) and the gravitational acceleration g=9,81 m/sec^2, the latitudinal friction coefficient can be written as:

$$\mu_e = \frac{V^2}{9{,}81\ \ R\ \ (3{,}6)^2} = \frac{V^2}{127\ \ R}$$

If the velocity term is taken from the equation, the critical velocity that causes the skid on an un-superelevated horizontal curve can be found as:

$$V_{skid} = 11{,}3\sqrt{\mu_e \cdot R}$$

The velocity that causes the rollover

The rollover effect of the F centrifugal force on a horizontal curve is compensated by the (Q.e/2) moment, which affects the point where the outer wheels of Q vehicle weight contact with the rolling surface.

So, the velocity that causes the rollover can be written using the moment equilibrium of the forces on point D as:

$$F\ \ h = Q.\frac{e}{2}$$

$$(\frac{Q}{g}\ \ \frac{v^2}{R})\ \ h = Q.\frac{e}{2}$$

$$v = \sqrt{\frac{e}{2}\ \ \frac{g\ \ R}{h}}$$

If the velocity of the vehicle is written as km/h (v=V/3,6) and the gravitational acceleration g=9,81 m/sec^2, the velocity that causes the rollover can be written as:

$$V_{ovr} = 8{,}0\sqrt{\frac{e \cdot R}{h}}$$

When the equation above is analyzed, it can be seen that the rollover velocity is a function of the radius of the horizontal curve and the vehicle size. On a curve with a specific radius, to show the effect of the vehicle size on the rollover, if the distance between the center of gravity of the vehicle and the contact point of the pavement with the vehicle wheel,

$$h = \frac{e}{2}\ \frac{g\ R}{v^2} \qquad h = e\ \frac{1}{\frac{2\ v^2}{g\ R}} \qquad h = \frac{e}{2\ \mu_e}$$

is obtained.

For example, if μ_e=0,50 is taken,

$$h = \frac{e}{2\ \ 0{,}50} \rightarrow\ h = e$$

can be obtained.

As known, the equilibrium states show the limit situations; on the h>e state, the rollover occurs. The critical velocity that causes the skid is lower than the critical velocity that causes the rollover. To show this condition, if the critical velocities are compared,

$$\frac{V_{skid}}{V_{ovr}} = \frac{11{,}3\sqrt{\mu_e \cdot R}}{8{,}0\sqrt{\frac{e \cdot R}{h}}} = 1{,}4\sqrt{\frac{\mu_e \cdot h}{e}}$$

is obtained.

For example, for a vehicle having the sizes of h=1,00 m ve e=1,60 m, traveling on a highway with μ_e=0,50 the ratio between the critical velocities is found as:

$$\frac{V_{skid}}{V_{ovr}} = 1{,}4\sqrt{\frac{0{,}50 \cdot 1{,}00}{1{,}60}} = 0{,}78 = \frac{78}{100}$$

If the vehicle height is decreased by 25%, or in other words, h=0,75 m is taken, this ratio becomes,

$$\frac{V_{skid}}{V_{ovr}} = 1{,}4\sqrt{\frac{0{,}50 \cdot 0{,}75}{1{,}60}} = 0{,}68 = \frac{68}{100}$$

Then if the e distance is decreased by 25%, or in other words, e=1,20 m is taken, this ratio becomes:

$$\frac{V_{skid}}{V_{ovr}} = 1{,}4\sqrt{\frac{0{,}50 \cdot 1{,}00}{1{,}20}} = 0{,}90 = \frac{90}{100}$$

Finally, if the e distance is increased by 25%, or in other words, e=2,00 is taken, this ratio becomes:

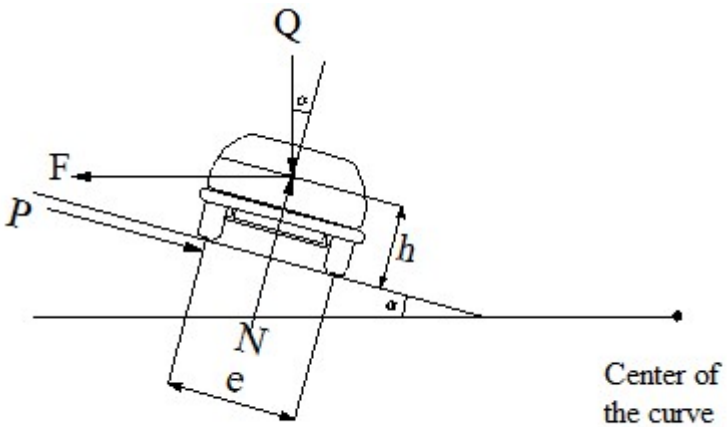
$$\frac{V_{skid}}{V_{ovr}} = 1{,}4\sqrt{\frac{0{,}50 \cdot 1{,}00}{2{,}00}} = 0{,}70 = \frac{70}{100}$$

These results show that the skid occurs more easily than the rollover and also show the effect of the change on vehicle sizes.

7.2.2 Stability on the road with a transverse slope

In practice, to counter the skid and rollover effects of the centrifugal force that affects the stability, the slope is given on the highway cross-section.

The slope, an example that can be seen in the figure, is given by changing the elevations of the highway axis or the highway sides and named the **superelevation**.



The critical velocities that cause the skid and rollover are also found using the equilibrium states for a superelevated curve like an un-superelevated curve.

For this reason, the centrifugal force and the vehicle weight's components parallel and perpendicular to the inclined plane have to be found.

The velocity that causes the skid

As the weight of the vehicle and the frictional force tries to resist the centrifugal force,

$$F \quad \cos\alpha = Q \quad \sin\alpha + P$$

equilibrium state can be written. As the P frictional force is the product of the normal force perpendicular to it and the friction coefficient, the equation can be written as:

$$F \quad \cos\alpha = Q \quad \sin\alpha + \mu_e \quad (Q \quad \cos\alpha + F \quad \sin\alpha)$$

If the expanded form of the F centrifugal force is written, the formula becomes,

$$\frac{Q}{g}\ \frac{v^2}{R}\ \cos\alpha = Q\ \ \sin\alpha + \mu_e\ \ (Q\ \ \cos\alpha + \frac{Q}{g}\ \frac{v^2}{R}\ \ \sin\alpha)$$

If each side of the equation is divided by cosα and the Q terms are canceled,

$$\frac{v^2}{g\ \ R}\ \ \frac{\cos\alpha}{\cos\alpha} = \frac{\sin\alpha}{\cos\alpha} + \mu_e\ \ (\frac{\cos\alpha}{\cos\alpha} + \frac{v^2}{g\ \ R}\ \ \frac{\sin\alpha}{\cos\alpha})$$

is obtained.

After the following cancellations are made on the equation, the critical skidding velocity can be found.

$$\frac{v^2}{g\ \ R} = tg\alpha + \mu_e\ \ (1 + \frac{v^2}{g\ \ R}\ \ tg\alpha)$$

$$\frac{v^2}{g\ \ R}(1\ \ \mu_e\ \ tg\alpha) = (tg\alpha + \mu_e)$$

$$v^2 = \frac{(tg\alpha + \mu_e)\ \ (g\ \ R)}{(1\ \ \mu_e\ \ tg\alpha)}$$

$$v = \sqrt{\frac{(tg\alpha + \mu_e)\ \ (g\ \ R)}{(1\ \ \mu_e\ \ tg\alpha)}}$$

If the velocity is taken as km/h (v=V/3,6) and the value of the gravitational acceleration is written as (g=9,81 m/sec^2), the **velocity that causes the skid on a superelevated condition** can be written as:

$$V_{skid} = 11{,}3\sqrt{\frac{R\cdot(tg\alpha + \mu_e)}{(1 - tg\alpha\cdot\mu_e)}}$$

If tgα=0 is written, the critical velocity value for non-superelevated curves can be obtained.

$$V_{skid} = 11{,}3\sqrt{\mu_e\cdot R}$$

The velocity that causes the rollover

The velocity that causes the rollover on a superelevated curve can be written using the moment equilibrium of the forces on point D. The equilibrium equation can be written as taking the moment of all forces according to point D as:

$$F \cos\alpha \; h = Q \sin\alpha \; h + Q \cos\alpha \; \frac{e}{2} + F \sin\alpha \; \frac{e}{2}$$

If the F centrifugal force is written on the equation,

$$\frac{Q}{g} \frac{v^2}{R} \cos\alpha \; h = Q \sin\alpha \; h + Q \cos\alpha \; \frac{e}{2} + \frac{Q}{g} \frac{v^2}{R} \sin\alpha \; \frac{e}{2}$$

If the both sides of the equation are divided by cosα and the velocity term is taken,

$$v = \sqrt{\frac{(tg\alpha \; h + \frac{e}{2}) \; (g \; R)}{(h \; tg\alpha \; \frac{e}{2})}}$$

is obtained.

If the velocity is taken as km/h (v=V/3,6) and the value of the gravitational acceleration (g=9,81 m/sn^2) is written, **the velocity that causes the rollover on a superelevated curve** can be written as:

$$V_{ovr} = 11{,}3 \sqrt{\frac{R \cdot (tg\alpha \cdot h + \frac{e}{2})}{(h - tg\alpha \cdot \frac{e}{2})}}$$

A highway is designed with respect to a selected project velocity. The minimum curve radius that would result in skid and rollover on a specific V_p project velocity can be calculated by the equations below:

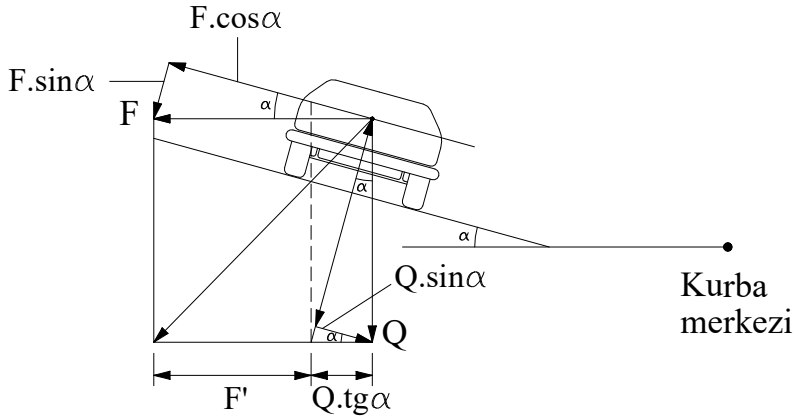
$$R_{\min(skid)} \geq \frac{V_p^2 \cdot (1 - tg\alpha \cdot \mu_e)}{127 \cdot (tg\alpha + \mu_e)}$$

$$R_{\min(ovr)} \geq \frac{V_p^2 \cdot (h - tg\alpha \cdot \frac{e}{2})}{127 \cdot (tg\alpha \cdot h + \frac{e}{2})}$$

7.3 Latitudinal acceleration and jerk

Vehicles moving on a superelevated horizontal curve are affected by the latitudinal acceleration caused by the centrifugal force.

The F' force that causes the latitudinal acceleration is seen in the figure.



The components of the centrifugal force and the weight of the vehicle are effective on the rise of the F' force. The first one is the parallel component of the centrifugal force to the superelevated road surface (F.cosα), and the second one is the parallel component of the weight of the vehicle to the superelevated road surface (Q.sinα). The counterpart of these forces on the horizontal is (Q.tgα). Normally, this force is expected to counter the centrifugal force, but an unbalanced force occurs, as seen in the figure. This force results in the latitudinal acceleration and can be written as:

$$F' = F \quad Q \quad tg\alpha$$

If the force terms and the weight term are written in terms of vehicle mass in this formula and p is used as the latitudinal acceleration

$$m \cdot p = m \cdot \frac{v^2}{R} - m \cdot g \cdot tg\alpha$$

is obtained. Here,

p : Latidunal acceleration (m/ sec^2)
m : Mass of the vehicle (kg)
v : Velocity of the vehicle (m/sn)
R : Horizontal curve radius (m)
g : Gravitational acceleration (m/sec^2)

As the tgα term shows the slope, or in other words, the superelevation, if tgα=q is taken, the equation becomes,

$$p = \frac{v^2}{R} \quad g \quad q$$

If the velocity is taken as km/h (v=V/3,6) and the value of the gravitational acceleration (g=9,81 m/sn^2) is written, the value of the latitudinal acceleration becomes,

$$p = \frac{V^2}{12{,}96.R} \quad 9{,}81 \quad q$$

If the latitudinal acceleration is positive (+), it affects through outer the curve or vice versa. The maximum value that does not disturb the travel comfort is accepted as 1,47 m/sec^2.

The change of acceleration for unit time is named the **jerk** and is used as a comfort criterion in highway design. The jerk is usually shown as p', and its unit is (m/sec^3). So it can be defined as:

$$p' = \frac{dp}{dt} = \frac{p}{t}$$

The jerk is especially important in the design of transition curves, which are used to decrease the sudden effect of centrifugal force. The t (sec) needed for a vehicle entering a horizontal curve with a velocity of v (m/sec) to pass the transition curve with a length of L_g (m) can be calculated as:

$$t = \frac{L_g}{v} = \frac{3,6 \quad L_g}{V}$$

If it is written on the jerk formula,

$$p' = \frac{V^2}{12,96.R} \quad \frac{V}{3,6 \quad L_g} \quad 9,81 \quad q \quad \frac{V}{3,6 \quad L_g} \qquad\qquad p' = \frac{V^3}{46,7.R \quad L_g} \quad \frac{2,7 \quad V \quad q}{L_g}$$

is obtained. Experimental studies show that under p'=0.3 m/sec^3, the jerk is not being felt at all; the p'=0.4 m/sec^3 value starts to disturb the travel comfort. As the length of the transition curve is being determined, the jerk value accepted is taken as p'=0,6 m/sec^3 to reflect the worst condition.

7.4 Minimum curve radius

The equilibrium state for superelevated curves can also be written as:

$$\frac{v^2}{g \quad R} = tg\alpha + \mu_e + (\mu_e \quad \frac{v^2}{g \quad R} \quad tg\alpha)$$

The third term on the right side of the equation can be neglectable because it is relatively small with respect to the others (especially the μ_e). If this neglected term is canceled, the equation becomes:

$$\frac{v^2}{g \quad R} = tg\alpha + \mu_e$$

If q superelevation is written instead of the slope tgα:

$$\frac{v^2}{g \quad R} = q + \mu_e$$

If the velocity is taken as km/h (v=V/3,6) and the value of the gravitational acceleration (g=9,81 m/sn^2) is written, the last form of the equation is:

$$\frac{V^2}{127 \quad R} = q + \mu_e$$

This situation clearly shows that the part of the centrifugal force is countered by the superelevation while the remaining part is countered by the latitudinal friction. Various methods have been developed to determine how the centrifugal force should be countered. Different countries have different implementations.

If R, the curve radius is taken from the equation, the minimum curve radius required for the project velocity can be written as:

$$R_{min} = \frac{V_p^2}{127 \quad (q + \mu_e)}$$

As stated before, the value of the latitudinal friction coefficient is related to many factors such as the type, smoothness, wetness, or dryness of the pavement; oldness or newness of tires; properties of tire treads, the pressure of the air inside the tires, the temperature of the tire while traveling; velocity of the vehicle. The value of the latitudinal friction coefficient is also high on dry and rough pavement. However, it is better to take low values due to the safety.

The maximum superelevation and latitudinal friction coefficients proposed by AASHTO and, accordingly, the minimum curve radius for various project velocities that correspond to the various highway types are given in the table. **The minimum curve radius is rounded up to the multiples of five.**

Minimum curve radius for various highway types

V_p (km/h	q_{max}	$\mu_{e,max}$	R_{min} (m)	V_p (km/h	q_{max}	$\mu_{e,max}$	R_{min} (m)
Urban				**Suburban roads**			
20	0,04	0,18	15	20	0,08	0,18	10
30	0,04	0,17	35	30	0,08	0,17	30
40	0,04	0,17	60	40	0,08	0,17	50
50	0,04	0,16	100	50	0,08	0,16	80
60	0,04	0,15	150	60	0,08	0,15	125
70	0,04	0,14	215	70	0,08	0,14	175
80	0,04	0,14	280	80	0,08	0,14	230
90	0,04	0,13	375	90	0,08	0,13	305
100	0,04	0,12	490	100	0,08	0,12	395
				110	0,08	0,11	500
				120	0,08	0,09	665
				130	0,08	0,08	830
Expressway				**Suburban rural**			
20	0,06	0,18	15	20	0,10	0,18	10
30	0,06	0,17	30	30	0,10	0,17	25
40	0,06	0,17	55	40	0,10	0,17	45
50	0,06	0,16	90	50	0,10	0,16	75
60	0,06	0,15	135	60	0,10	0,15	115
70	0,06	0,14	195	70	0,10	0,14	160
80	0,06	0,14	250	80	0,10	0,14	210
90	0,06	0,13	335	90	0,10	0,13	275
100	0,06	0,12	435	100	0,10	0,12	360
110	0,06	0,11	560	110	0,10	0,11	455
120	0,06	0,09	755	120	0,10	0,09	595
130	0,06	0,08	950	130	0,10	0,08	740

7.5 Superelevation

The latitudinal slope given to the highway cross-section to counter the skid and the rollover effects of centrifugal force while traveling on a horizontal curve with a specific velocity is named **superelevation**. An important element contributing to the stability is the latitudinal friction. If an ideal situation with no latitudinal friction is assumed, the whole centrifugal force is to be countered by the superelevation. When μ_e=0 is written,

$$\frac{V^2}{127\ R} = q$$

is obtained. The latitudinal slope found in this situation is named the **theoretical superelevation**. This means that, on a curve without the effect of the latitudinal friction, theoretically, the superelevation needed to counter the centrifugal force is related to the velocity of the vehicle and the curve radius. Naturally, the value of the latitudinal acceleration would be zero. The same result can be reached by taking p=0. The theoretical superelevation on the highway designed with the project velocity of V_p is calculated as:

$$q_{theo} = \frac{V_p^2}{127 \cdot R} = 0{,}00786 \cdot \frac{V_p^2}{R}$$

If the superelevation is used as the percentage, it can be written as:

$$q_{teo} = 0{,}786\ \frac{V_p^2}{R}$$

However, the theoretical superelevation is not directly applied because its value is high. When the vehicles (especially the heavy vehicles) are traveling with velocities lower than the

specified project velocity or during their stops on horizontal curves, for the slopes found with the theoretical superelevation formula, they can lose their stability and skid or rollover through the center of the curve. Moments that arise, especially on over or unbalanced loading, ease the skid or rollover because the centripetal force created by the high latitudinal slope can be higher than the centrifugal force in these conditions. As a result, a q_{max} has to be determined as a limit value in use.

On the other hand, in real-life conditions, friction may be low, but its effects can not be neglected. Thus, the contribution of the latitudinal friction to counter the effect of the centrifugal force can not be neglected. So the problem of what amount of the centrifugal force is to be countered by the superelevation and what amount of it is to be countered by the latitudinal friction arises. There are several approaches to this problem. In some countries, half of the effect of the centrifugal force is countered by the superelevation, while the remaining half is by the latitudinal friction. The superelevation calculated this way is named as the **practical superelevation**. In this condition, the superelevation formula becomes:

$$q_{pra} = 0{,}786 \cdot \frac{V_p^2}{R} \cdot 0{,}50 = 0{,}393 \cdot \frac{V_p^2}{R}$$

The practical superelevation applied in Turkey is acquired by accepting that 56% of the effect of centrifugal force is countered by the superelevation and the remaining 44% by the latitudinal friction. So the equation used on design calculations can be written as:

$$q_{pra} = 0{,}786 \cdot \frac{V_p^2}{R} \cdot 0{,}56 = 0{,}443 \cdot \frac{V_p^2}{R}$$

The upper limit of the superelevation varies with respect to the highway type and, accordingly, the project velocity, usually between 4%-10%. As the velocities are low in urban highways, the maximum superelevation is also low. On suburban highways, taking the type of highway, snow, and icing conditions in winter into account, 8%-10% values can be used. On particular conditions, q_{max}=12% can also be used.

According to the drainage needs and being a suburban highway, the superelevation corresponding to the crown slope between 1,5%-3% is named as the minimum superelevation (q_{min}). The most common crown slope on application is q_{min} = 2%.

7.6 The application of the superelevation

As already known, a highway design is made proper to the project standards determined with respect to the selected highway class. Project standards are important on defining the limit values to be respected during the design. In application, superelevation starts from a point on the alignment before a specific distance from the start of the curve and reaches its maximum value on a specific point on the curve. Making this change is named the application of the superelevation. The application of the superelevation is also related to some standards.

The maximum superelevation q_{max}, the minimum superelevation which is also named as the crown slope q_{min}, and the minimum superelevation implementation length $L_{d,min}$ are the major ones. On highway sections with high curve radii or on low-standard highways, additional highway components to increase comfort (transition curves) may not be used when entering a horizontal curve from an alignment. In this case, superelevation on a curve without the transition curve is applied. First of all, this situation will be emphasized. The application of superelevation on a curve with a transition curve has some differences, which will be emphasized later on.

7.6.1 Determination of the maximum superelevation percentage

The maximum superelevation percentage is found by using the formula below for the selected project velocity and the applied curve radius, as already stated.

$$q = 0{,}443 \quad \frac{V_p^2}{R}$$

Here, $q \le q_{max}$

constraint has to be complied with.

If it is impossible to comply with this constraint or, in other words, if $q > q_{max}$

Then $q = q_{max}$

is applied, or either the curve radius is increased without changing the project velocity or the project velocity is decreased without changing the curve radius.

The velocity found this way is named as the restricted velocity (V_r), and traffic signs are placed on this section of the highway to warn the drivers in application.

Sample problem

A horizontal curve with a radius of R=600 m is designed for a highway with the project velocity of V_p = 90 km/h, a crown slope of q_{min}=2% and a maximum superelevation of q_{max}=8%. Check the suitability of the curve radius with respect to the superelevation.

Solution:

The suitability of the curve radius can be checked by two methods.

The first method is the calculation of the maximum superelevation that the curve properties require. The value found is compared with the value of q_{max}=8%; if it is smaller or equal to it, then it is suitable. If it is bigger, either velocity is restricted, or the curve radius is increased. As,

$$q = 0{,}443 \quad \frac{V_p^2}{R} = 0{,}443 \quad \frac{90^2}{600} = 6{,}0\% < q_{max} = 8\%$$

the curve radius selected is suitable.

The second method is the calculation of the minimum curve radius by taking maximum superelevation q_{max}=%8 using the practical superelevation formula. The minimum curve radius found this way is compared with the given curve radius. If it is smaller or equal to it, then it is suitable. If it is bigger, either velocity is restricted, or the curve radius is increased. As,

$$q = 0{,}443 \quad \frac{V_p^2}{R} \rightarrow \quad 8 = 0{,}443 \quad \frac{90^2}{R_{min}} \rightarrow \quad R_{min} = 0{,}443 \quad \frac{90^2}{8} = 448{,}5\,\text{m} \quad 450\,\text{m} < 600\,\text{m}$$

the curve radius selected is suitable.

Sample problem

A horizontal curve with a radius of R=200 m is designed for a highway with a project velocity of V_p = 80 km/h, a crown slope of q_{min}=2%, and a maximum superelevation of q_{max}=8%. Check the suitability of the curve radius with respect to the superelevation.

Solution :

First, the superelevation is checked. As,

$$q = 0{,}443 \cdot \frac{V_p^2}{R} = 0{,}443 \cdot \frac{80^2}{200} = \%14{,}2 > q_{max} = \%8$$

either the velocity must be restricted, or the curve radius must be increased. If the velocity is restricted,

$$8 = 0{,}443 \cdot \frac{V_r^2}{200} \rightarrow V_r = 60{,}10 \cong 60 km/h$$

is taken, and the curve radius of R=200 m is applied. If the project velocity is preserved, the curve radius must be changed. If there is enough distance to place the curve, the curve radius can be chosen as:

$$R_{min} = 0{,}443 \cdot \frac{V_p^2}{q} = 0{,}443 \cdot \frac{80^2}{8} = 354{,}4m \cong 355m$$

On the other hand, if the superelevation needed is found as $q<q_{min}$, there's no need to apply superelevation on the curve with the given radius.

Change of superelevation with respect to the project velocity and the curve radius

Superelevation q (%)	Project velocity (Vp)									
Curve radius	30	40	50	60	70	80	90	100	110	120
R (m)	(km/h)	(km/h)	(km/h)	(km/h)	(km/h)	(km/h)	(km/h)	(km/h)	(km/h)	(km/h)
3.250	-	-	-	-	-	-	-	-	-	2,0
3.000	-	-	-	-	-	-	-	-	-	2,1
2.700	-	-	-	-	-	-	-	-	2,0	2,4
2.500	-	-	-	-	-	-	-	-	2,1	2,5
2.200	-	-	-	-	-	-	-	2,0	2,4	2,9
2.150	-	-	-	-	-	-	-	2,1	2,5	3,0
2.000	-	-	-	-	-	-	-	2,2	2,7	3,2
1.750	-	-	-	-	-	-	2,0	2,5	3,0	3,6
1.450	-	-	-	-	-	-	2,5	3,0	3,7	4,4
1.400	-	-	-	-	-	2,0	2,6	3,1	3,8	4,5
1.150	-	-	-	-	-	2,5	3,1	3,8	4,6	5,5
1.100	-	-	-	-	2,0	2,6	3,2	4,0	4,9	5,8
1.000	-	-	-	-	2,2	2,8	3,6	4,4	5,3	6,3
880	-	-	-	-	2,5	3,2	4,1	5,0	6,1	7,2
800	-	-	-	2,0	2,7	3,5	4,5	5,5	6,7	7,9
635	-	-	-	2,5	3,4	4,4	5,6	6,9	8,4	10,0
550	-	-	2,0	2,9	3,9	5,1	6,5	8,0	9,7	-
535	-	-	2,1	3,0	4,0	5,3	6,7	8,2	10,0	-
500	-	-	2,2	3,2	4,3	5,6	7,1	8,8	-	-
445	-	-	2,5	3,6	4,9	6,3	8,0	9,9	-	-
400	-	-	2,8	4,0	5,4	7,1	8,9	-	-	-
360	-	2,0	3,1	4,4	6,0	7,8	9,9	-	-	-
300	-	2,4	3,7	5,3	7,2	9,4	-	-	-	-
285	-	2,5	3,9	5,6	7,6	9,9	-	-	-	-
220	-	3,2	5,0	7,2	9,8	-	-	-	-	-
200	2,0	3,5	5,5	7,9	-	-	-	-	-	-
160	2,5	4,4	6,9	9,9	-	-	-	-	-	-
150	2,6	4,7	7,3	-	-	-	-	-	-	-
115	3,5	6,1	9,6	-	-	-	-	-	-	-
100	4,0	7,1	-	-	-	-	-	-	-	-
80	5,0	8,8	-	-	-	-	-	-	-	-
75	5,3	9,4	-	-	-	-	-	-	-	-
60	6,6	-	-	-	-	-	-	-	-	-
50	7,9	-	-	-	-	-	-	-	-	-
40	9,9	-	-	-	-	-	-	-	-	-

The minimum curve radii (Rmin) (m) for q_{max}=8%

Project velocity (km/h)	30	40	50	60	70	80	90	100	110	120
Calculated value	49,84	88,60	138,44	199,35	271,34	354,40	448,54	553,75	670,04	797,40
Rounded value	50	90	140	200	275	355	450	555	675	800

The restricted velocities (V_r) for q_{max} =%8

	Project velocity(V_p)									
Curve radius R (m)	30 (km/h)	40 (km/h)	50 (km/h)	60 (km/h)	70 (km/h)	80 (km/h)	90 (km/h)	100 (km/h)	110 (km/h)	120 (km/h)
800	-	-	-	-	-	-	-	-	-	-
635	-	-	-	-	-	-	-	-	107	107
550	-	-	-	-	-	-	-	-	100	100
535	-	-	-	-	-	-	-	98	98	98
500	-	-	-	-	-	-	-	95	95	95
445	-	-	-	-	-	-	-	90	90	90
400	-	-	-	-	-	-	85	85	85	85
360	-	-	-	-	-	-	81	81	81	81
300	-	-	-	-	-	74	74	74	74	74
285	-	-	-	-	-	72	72	72	72	72
220	-	-	-	-	63	63	63	63	63	63
200	-	-	-	-	60	60	60	60	60	60
160	-	-	-	54	54	54	54	54	54	54
150	-	-	-	52	52	52	52	52	52	52
115	-	-	46	46	46	46	46	46	46	46
100	-	-	42	42	42	42	42	42	42	42
80	-	38	38	38	38	38	38	38	38	38
75	-	37	37	37	37	37	37	37	37	37
60	-	33	33	33	33	33	33	33	33	33
50	-	30	30	30	30	30	30	30	30	30
40	27	27	27	27	27	27	27	27	27	27

7.6.2 The determination of the superelevation application length

The superelevation does not reach its highest value in an instant. To provide a safe and comfortable transition, a distance is needed. This distance is called as the superelevation application length and can be calculated as:

$$L_d = 0{,}0354 \cdot \frac{V_p^3}{R}$$

This value found has to provide the constraint of

$$L_d \geq L_{se,\min} = 45m$$

If it does not, $L_d = L_{min}$ =45 m is taken for calculations.

As a vehicle moving on an alignment enters a horizontal curve, because of the effect of the centrifugal force, its stability is corrupted, and its passengers start to feel discomfort. The precautions to counter this effect have to be taken at the entrance of the curve. For this reason, the crown slope must change into the superelevation or a single slope when the curve is entered.

On application, superelevating starts from a specific distance before the curve entrance and on the alignment, and the superelevation reaches its maximum value at a specific point on the curve. As a result of the experiments, superelevation was applied between $2/3L_d$ before the start of the curve (PC) and $1/3L_d$ after the start of the curve. In other words, under normal

conditions, the superelevation application starts at least 30 m before the PC and reaches its maximum value at least 15 m after.

The superelevation increases linearly during this application. Because of the linear increase, the superelevation of any point can be calculated easily.

Sample problem

A horizontal curve with a radius of R=500 m is designed for a highway with a project velocity of Vp = 90 km/h, a crown slope of qmin=2%, and a maximum superelevation of q_{max}=8%. Check the suitability of the curve radius with respect to the superelevation and find the superelevation application length.

Solution:

Primarily the maximum value of the superelevation with respect to the given curve properties is calculated. As

$$q = 0{,}443 \quad \frac{V_p^2}{R} = 0{,}443 \quad \frac{90^2}{500} = \%7{,}1 < q_{max} = \%8$$

the radius of the curve is suitable. The superelevation application length is found as:

$$L_d = 0{,}0354 \cdot \frac{V_p^3}{R} = 0{,}0354 \cdot \frac{90^3}{500} = 51{,}6 \text{ m}$$

As $L_d > L_{d,min}$=45 m, this length can be applied.

Sample problem

A horizontal curve with a radius of R=300 m is designed for a highway with a project velocity of Vp = 70 km/h, a crown slope of qmin=2%, and a maximum superelevation of q_{max}=8%. Check the suitability of the curve radius with respect to the superelevation and find the superelevation application length.

Solution:

As ,

$$q = 0{,}443 \cdot \frac{V_p^2}{R} = 0{,}443 \cdot \frac{70^2}{300} = \%7{,}2 < q_{max} = \%8$$

the curve radius is suitable. The superelevation application length can be calculated as:

$$L_d = 0{,}0354 \quad \frac{V_p^3}{R} = 0{,}0354 \quad \frac{70^3}{300} = 40{,}5 \text{ m}$$

As $L_d < L_{d,min}$=45 m, L_d=45 m is applied.

The change of the superelevation application length (L_d) (q_{max}=8%)

Superelevation dist. L_{se} (m)	Project velocity(V_p)									
Curve radius R (m)	30 (km/h)	40 (km/h)	50 (km/h)	60 (km/h)	70 (km/h)	80 (km/h)	90 (km/h)	100 (km/h)	110 (km/h)	120 (km/h)
3.250	-	-	-	-	-	-	-	-	-	45,0
3.000	-	-	-	-	-	-	-	-	-	45,0
2.700	-	-	-	-	-	-	-	-	45,0	45,0
2.500	-	-	-	-	-	-	-	-	45,0	45,0
2.200	-	-	-	-	-	-	-	45,0	45,0	45,0
2.150	-	-	-	-	-	-	-	45,0	45,0	45,0
2.000	-	-	-	-	-	-	-	45,0	45,0	45,0
1.750	-	-	-	-	-	-	45,0	45,0	45,0	45,0
1.450	-	-	-	-	-	-	45,0	45,0	45,0	45,0
1.400	-	-	-	-	-	45,0	45,0	45,0	45,0	45,0
1.150	-	-	-	-	-	45,0	45,0	45,0	45,0	53,2
1.100	-	-	-	-	45,0	45,0	45,0	45,0	45,0	55,6
1.000	-	-	-	-	45,0	45,0	45,0	45,0	47,1	61,2
880	-	-	-	-	45,0	45,0	45,0	45,0	53,5	69,5
800	-	-	-	45,0	45,0	45,0	45,0	45,0	58,9	76,5
635	-	-	-	45,0	45,0	45,0	45,0	55,7	-	-
550	-	-	45,0	45,0	45,0	45,0	46,9	-	-	-
535	-	-	45,0	45,0	45,0	45,0	48,2	-	-	-
500	-	-	45,0	45,0	45,0	45,0	51,6	-	-	-
445	-	-	45,0	45,0	45,0	45,0	-	-	-	-
400	-	-	45,0	45,0	45,0	45,3	-	-	-	-
360	-	45,0	45,0	45,0	45,0	50,3	-	-	-	-
300	-	45,0	45,0	45,0	45,0	-	-	-	-	-
285	-	45,0	45,0	45,0	45,0	-	-	-	-	-
220	-	45,0	45,0	45,0	-	-	-	-	-	-
200	45,0	45,0	45,0	45,0	-	-	-	-	-	-
160	45,0	45,0	45,0	-	-	-	-	-	-	-
150	45,0	45,0	45,0	-	-	-	-	-	-	-
115	45,0	45,0	-	-	-	-	-	-	-	-
100	45,0	45,0	-	-	-	-	-	-	-	-
80	45,0	-	-	-	-	-	-	-	-	-
75	45,0	-	-	-	-	-	-	-	-	-
60	45,0	-	-	-	-	-	-	-	-	-
50	45,0	-	-	-	-	-	-	-	-	-

Sample problem

A horizontal curve with a radius of R=200 m is designed for a highway with a project velocity of Vp = 80 km/h, a crown slope of qmin=2%, and a maximum superelevation of q_{max}=8%. Find the superelevation application length (it's impossible to change the curve radius).

Solution:

Primarily, the superelevation is checked. As

$$q = 0{,}443 \cdot \frac{V_p^2}{R} = 0{,}443 \cdot \frac{80^2}{200} = \%14{,}2 > q_{\max} = \%8$$

the velocity has to be restricted. The restricted velocity can be found as:

$$8 = 0{,}443 \cdot \frac{V_r^2}{200} \rightarrow V_r = 60{,}10 \cong 60 km/h$$

Then by taking $V_p = V_c$, superelevation application length is calculated as:

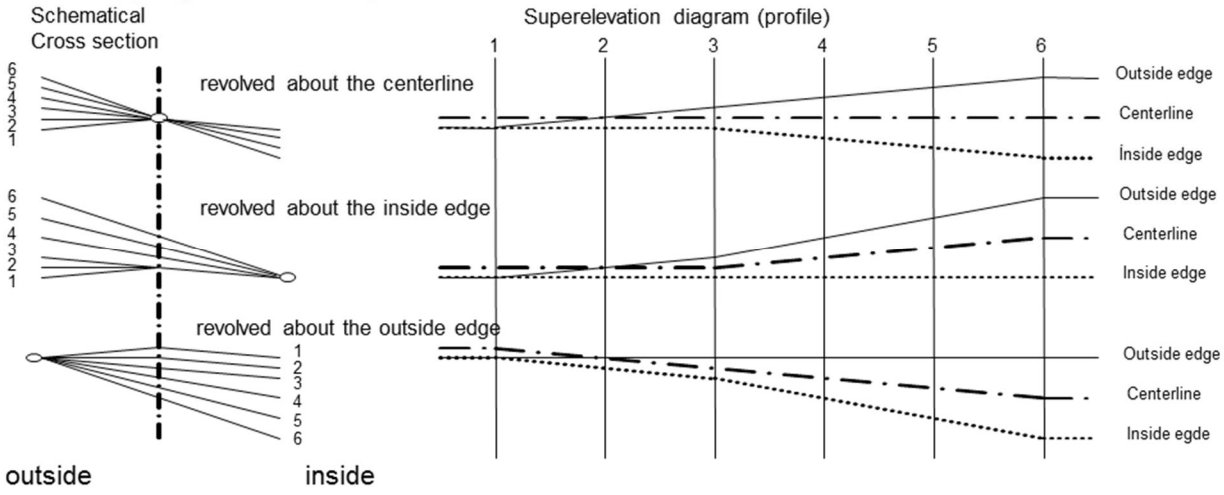
$$L_d = 0{,}0354 \cdot \frac{V_r^3}{R} = 0{,}0354 \cdot \frac{60^3}{200} = 38{,}2m$$

As $L_d < L_{d,min} = 45$ m, $L_d = 45$ m is taken for application. As shown in the example, after the control of the superelevation constraint, the velocity restriction also has to be checked and applied if needed.

7.6 Implementing the change of superelevation through the length

Superelevation can be implemented by 3 different methods according to the position of the highway axis to the inner edge and outer edge of the highway with respect to the curve center. These are:

1. By keeping the elevation of the highway axis constant while decreasing the elevation of the inner edge and increasing the elevation of the outer edge (**implementing superelevation by keeping the axis constant**),
2. By keeping the elevation of the inner edge of the highway constant while increasing the elevations of the axis and the outer edge (**implementing superelevation by keeping the inner edge constant**),
3. By keeping the elevation of the outer edge of the highway constant while increasing the elevations of the axis and the inner edge (**implementing superelevation by keeping the outer edge constant**),



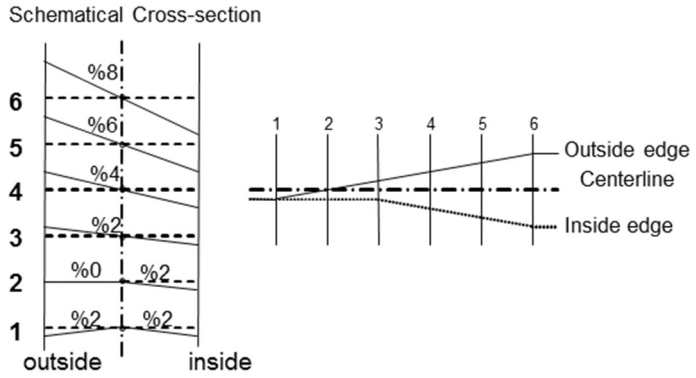
The main advantage of the first method on which the highway revolves around the centerline is there is no need to change the elevations on the profile. On the other two methods on which the highway revolves around the edges, superelevations of the profile have to be re-calculated because the elevations of the centerline change. If the highway revolves around the inner edge, elevations of the centerline increase; if the highway revolves around the outer edge, they decrease.

Another important difference comes up about the drainage. If the highway revolves around the centerline or the outer edge, a significant cavitation occurs on the inner edge of the highway. Cavitation may complicate the drainage of the surface water.

7.6.3.1 Implementing superelevation by keeping the axis constant

In this method, the crown slope started to be changed $2/3L_d$ before the curve. Change is made by revolving the edges around the centerline (Cross section 1). At first, the elevation of the outer edge is increased to reach the elevation of the centerline (Cross section 2). In this case, the latitudinal slope of the outer side of the platform becomes zero. The superelevation of the inner side is still at the crown slope.

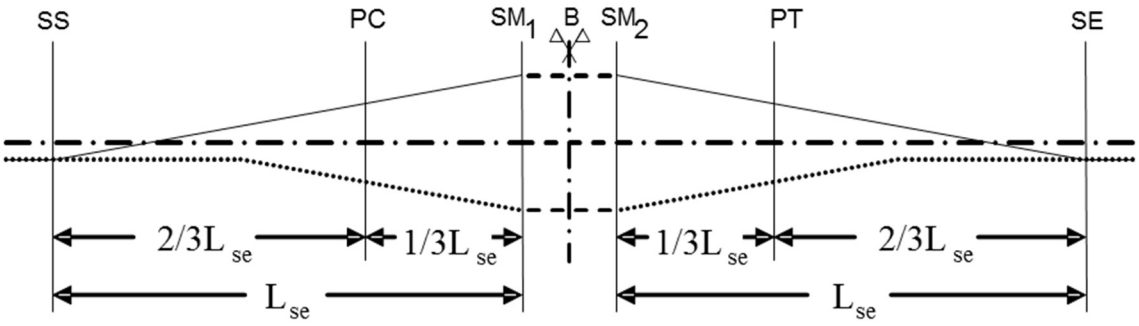
Then, the elevation of the outer edge continues to rise and reaches the same slope as the inner edge (Cross section 3). In this case, the absolute value of the superelevation heights of the inner and the outer edges are the same with respect to the centerline, inner edge below, and outer edge over it. The highway platform reaching the single slope on this section continues to revolve around the centerline. On each revolvement, the elevation of the inner edge is increased while the elevation of the outer edge is decreased (Cross sections 4 and 5). The amounts of change are the same with respect to the centerline. While revolving around the centerline, the horizontal curve is entered. In other words, a rolling surface with a single slope is reached to counter the negative effects of the centrifugal force. $1/3L_d$ after the start of the horizontal curve, the maximum amount of the superelevation is reached and the revolvement is finalized (Cross section 6). After this point, superelevation is applied constantly through the circle arc that forms the horizontal curve.



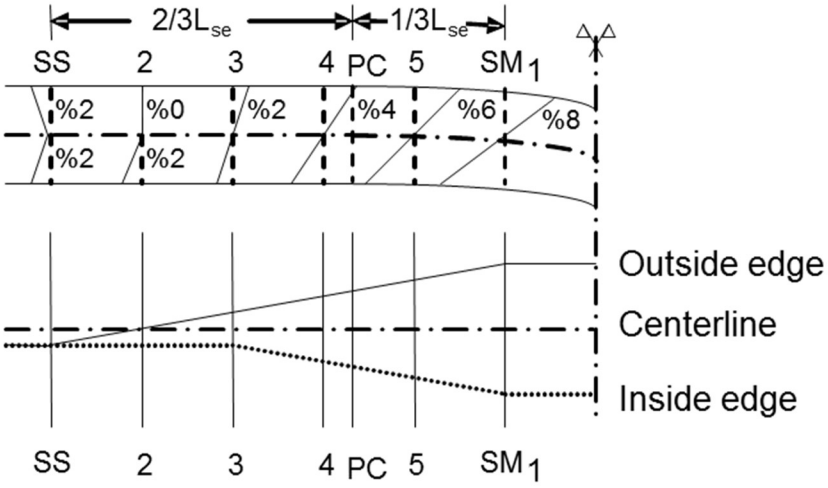
The same processes are followed inversely on the curve exit. In other words, the revolvement starts in the opposite direction $1/3L_d$ before the PT. This point is the last point with the highest superelevation. When the PT is reached, some amount of superelevation is still present to counter the effects of the centrifugal force. The superelevation is ended by reaching the crown slope $2/3L_d$ after the PT.

The point that the superelevation starts to change is noted by (SS), while the first point that the superelevation has the maximum amount as (SM_1), symmetrically the last point that the superelevation has the maximum amount as (SM_2) and the point that the superelevation

returns to the crown slope as (SE). An example of the usage of the notation is given in the figure below.



The superelevation plan and the superelevation diagram for a horizontal curve are given in the figure below.



A superelevation diagram is also a profile drawn through the superelevation application length, and it shows the superelevation heights with respect to the elevation of the centerline. The elevation increase or decrease amount can be calculated as:

$$h = q \quad \frac{b}{2}$$

taking b as the platform width. If the value obtained is negative (-), it is decreased with respect to the elevation of the centerline; if it is positive (+), it is increased with respect to the elevation of the centerline. In the same way, to calculate the minimum and the maximum superelevation heights,

$$h_{min} = q_{min} \cdot \frac{b}{2} \qquad h_{max} = q_{max} \cdot \frac{b}{2}$$

equations are used. If the horizontal curve is widened or constricted, the platform widths in the equation have to be corrected, and calculations have to be made with respect to this value. On application, the superelevation percentages and the superelevation amounts have to be calculated for cross-sections that fall inside the superelevation application length, as shown in the table. Using these heights, the red elevations on the centerline and the edges can also be calculated.

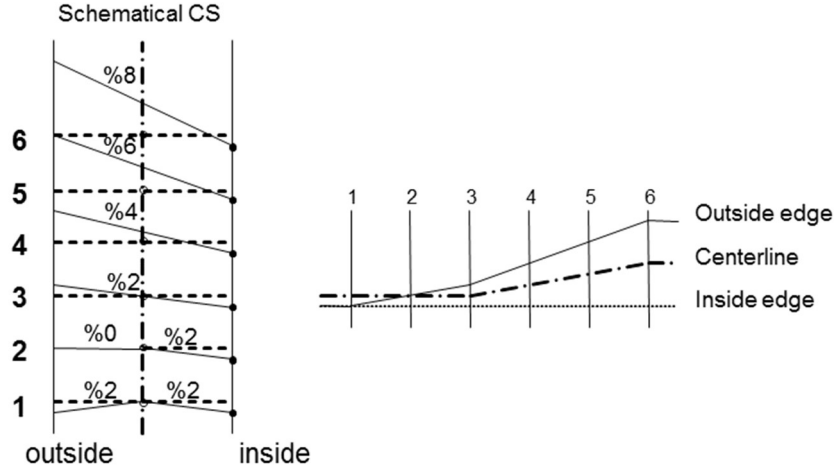
The change of the elevation on a highway revolved around the centerline

CS Nr.	Distance to Start (m)	Superelevation(%)		Elevation change wrt the centerline on SS		
		Outside E.	Inside E.	Outside E. (m)	Centerline (m)	Inside E. (m)
1(SS)	0	-2,0	-2,0	-0,10	0,00	-0,10
2	9	0,0	-2,0	0,00	0,00	-0,10
3	18	2,0	-2,0	0,10	0,00	-0,10
4	27	4,0	-4,0	0,20	0,00	-0,20
PC	30	4,7	-4,7	0,235	0,00	-0,235
5	36	6,0	-6,0	0,30	0,00	-0,30
6(SM_1)	45	8,0	-8,0	0,40	0,00	-0,40

V_p=80 km/h, R=200 m, b=10 m, q_{min}=%2, q_{max} =%8, L_d=45 m

7.6.3.2 Implementing superelevation by keeping the inner edge constant

In this method, the processes followed up to reaching a single slope (Cross-section 3) are the same as the first method. In other words, up to this point, the outer edge revolves around the centerline, and there's no change in the elevation of the inner edge. After this point, the platform revolves around the constant inner edge, and the superelevation amount is increased (Cross-sections 4 and 5). As seen in the figure below, while the elevation of the outer edge is increased, the elevation of the centerline also increases. An example of the amount of change is given in the table for the project criteria and the curve radius. The superelevation reaches the maximum value $1/3L_d$ after the entrance of the horizontal curve (Cross-section 6) and continues this way until $1/3L_d$ before the PT. The processes are the same as the previous method. As the elevations of the inner edge does not change, the best method with respect to the drainage is this one.



The change of the elevation on a highway revolved around the inner edge

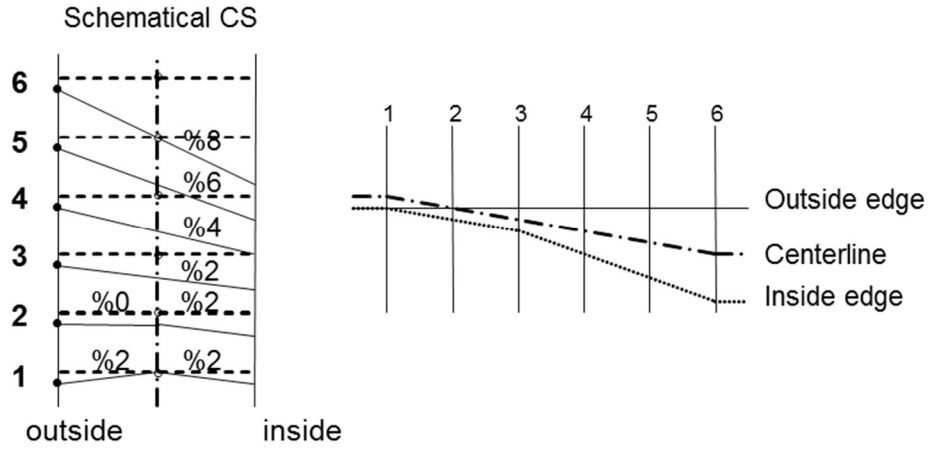
CS Nr.	Distance to start(m)	Superelevation(%)		Elevation change wrt the centerline on SS		
		Outside E	Inside E.	Outside E.(m)	Centerline (m)	Inside E. (m)
1(SS)	0	-2,0	-2,0	-0,10	0,00	-0,10
2	9	0,0	-2,0	0,00	0,00	-0,10
3	18	2,0	-2,0	0,10	0,00	-0,10
4	27	4,0	-4,0	0,30	0,10	-0,10
PC	30	4,7	-4,7	0,37	0,135	-0,10
5	36	6,0	-6,0	0,50	0,20	-0,10
6(SM_1)	45	8,0	-8,0	0,70	0,30	-0,10

V_p=80 km/h, R=200 m, b=10 m, q_{min}=%2, q_{max} =%8, L_{se} =45 m

7.6.3.3 Implementing superelevation by keeping the outer edge constant

In this method, the crown slope starts to change $2/3L_{se}$ before the entrance of the horizontal curve (Cross-section 1). As the elevation of the outer edge remains constant, the elevation of the centerline and the inner edge have to be decreased together. The slope of the outer side of the platform is decreased to zero by decreasing the elevation of the centerline. During this process, the elevations of the inner edge decrease by the same amount. However, the crown slope of this section does not change (Cross-section 2). Then, the revolvement continues to reach the platform and reaches a single slope (Cross-section 3). In this case, the elevations of the inner edge continue to decrease parallelly to the decrease of the elevations of the centerline. As the single slope is obtained, the revolvement and the elevation decrease are continued until the maximum superelevation amount is reached $1/3L_{se}$ after the PC (Cross-sections 4, 5, and 6). Similar to the previous methods, the same processes are followed for the other side of the curve to finalize the application.

The change in the elevations of the centerline and the edges is shown in the table. The elevations of the inner edge are very low compared to the first method. This situation is the main cause of the drainage problem.



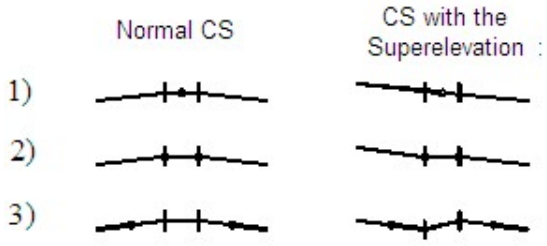
The change of the elevation on a highway revolved around the outer edge

CS Nr.	Distance to start(m)	Superelevation(%)		Elevation change wrt the centerline on SS		
		Outside E	Inside E.	Outside E.(m)	Centerline (m)	Inside E. (m)
1(DB)	0	-2,0	-2,0	-0,10	0,00	-0,10
2	9	0,0	-2,0	-0,10	-0,10	-0,20
3	18	2,0	-2,0	-0,10	-0,20	-0,30
4	27	4,0	-4,0	-0,10	-0,30	-0,50
ò	30	4,7	-4,7	-0,10	-0,335	-0,57
5	36	6,0	-6,0	-0,10	-0,40	-0,70
6(DM_i)	45	8,0	-8,0	-0,10	-0,50	-0,90

V_p=80 km/h, R=200 m, b=10 m, q_{min}=%2, q_{max} =%8, L_{se}=45 m

7.6.4 Application of the superelevation on double platform highways

Double platform highways are roads where the traffic in one direction is separated from the traffic in the other direction with a central refuge or another physical obstacle. The superelevation can be applied to various types with respect to the functions and properties of the central refuge. Thus, the revolvement point gains importance.



In the first example, on a highway with a crown slope on refuge edges, the center of the refuge is constant, and the platforms revolve around this point to apply superelevation. In this case, the slope of the refuge has the same direction with the superelevation. In the second example, the edges on each side of the platform remain constant for each direction. In this case, one direction revolves around the inner edge while the other one revolves around the outer. The latitudinal slopes have to be parallel for each platform. The slope of the refuge is zero.

In the third example, each platform is revolved around its own centerline. As a result of this application, the slope of the refuge is inverse to the slope of the platform. In any case, the drainage of the surface water has to be considered.